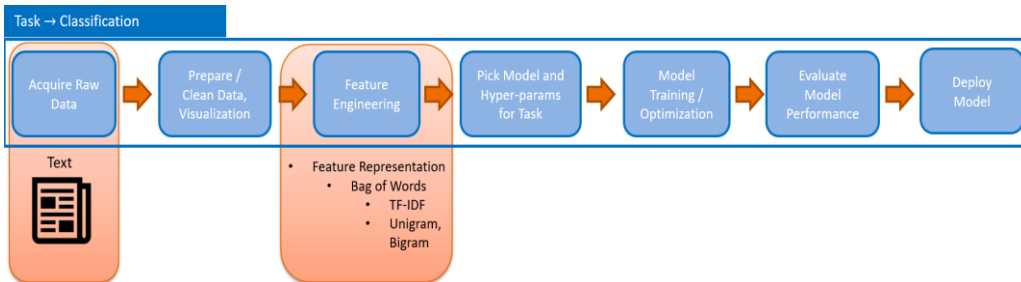
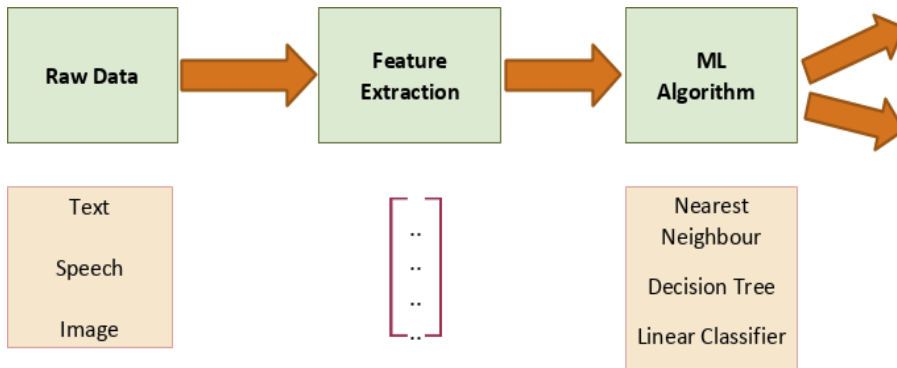


Focus for this lecture



Introduction

Pipeline



Feature Engineering

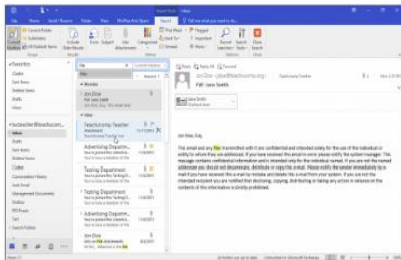
Representation

Three Generations of Features or Representations

1. Intuition driven representations (hand-crafted)
2. Representations that are derived based on Statistics, Signal Processing etc.
3. Representations that are learned

Problem: Classify Textual Content

Emails



Web Page



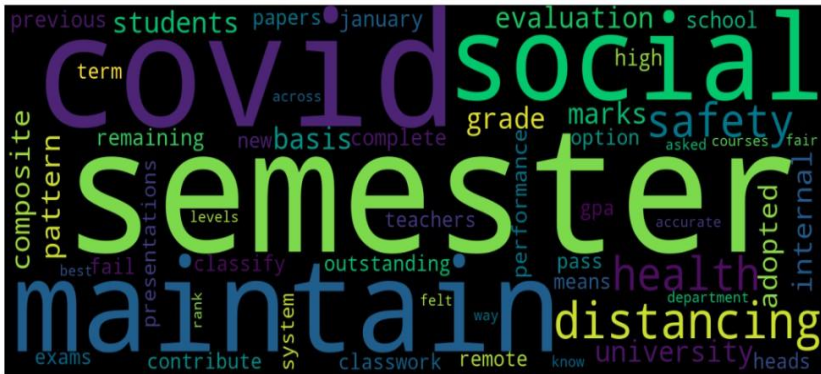
- Problem: Classify into one of C classes.
- Eg. Email: Spam vs. Non-spam, Professional vs. Personal, Web Page: Movies vs. Sports Vs Politics

What Do We Mean by Representing Text?

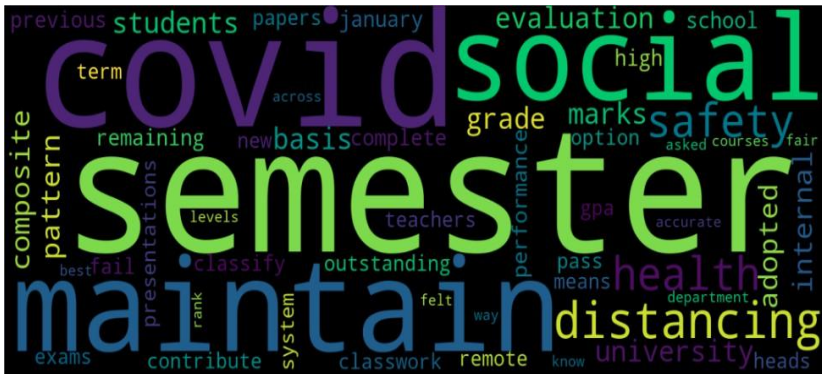
- ▶ Representations for words?
- ▶ Representations for phrases?
- ▶ Representations for sentences?
- ▶ Representations for documents?

Bag of Words - Text Domain - Motivation

- ▶ Word cloud: The size of the word indicates how often the word occurs in the document

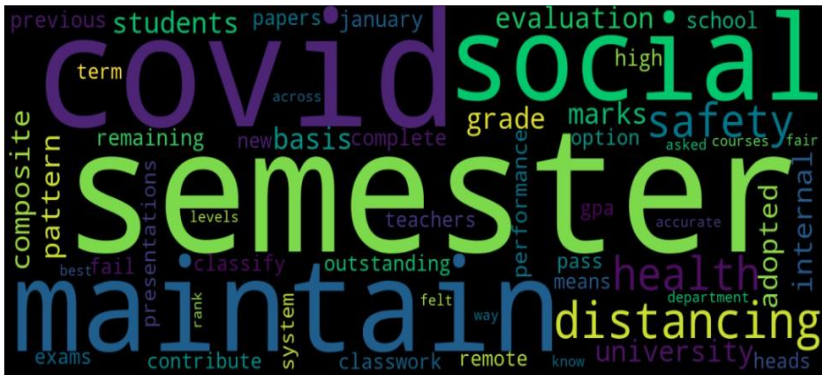


- What is the document talking about?



Bag of Words - Text Domain - Motivation

- Orderless document representation; frequencies of words from a dictionary



Problem Statement

Problem Statement

- ▶ **Input:** Covid-19 education grading new system

Problem Statement

- ▶ **Input:** Covid-19 education grading new system

The HRD ministry said that, In case the situation does not appear to be normal this semester in view of Covid19, to maintain social distancing, safety, and health of the university students grading could be a composite of 50% marks on the basis of the pattern of internal evaluation adopted by the university and the remaining 50% marks on the basis of performance in the previous semester.

Problem Statement

- **Input:** Covid-19 education grading new system

The HRD ministry said that, In case the situation does not appear to be normal this semester in view of Covid19, to maintain High school students will have until health January 2021 to complete any could outstanding classwork for grades the be during Covid-19, which classify evalu into remote term papers, the re presentations and exams. They will perform have the option to classify their class work on a pass/fail basis, which means their grade would not contribute to their GPA.

Problem Statement

- **Input:** Covid-19 education grading new system

The HRD ministry said that, In case the situation does not appear to be normal this semester in view of Covid19, to

maintain High school students will have until January 2021 to complete any outstanding classwork for grades

Under the new system, university teachers and their heads of department, will be asked to grade and rank their students. It is felt that as teachers know their students best, this is the most fair and accurate way of evaluation of a student's performance across the course of their A-Levels.

Problem Statement

- **Input:** Covid-19 education grading new system
- **Task:** Identify the countries from the article.

The HRDministry said that, In case the situation does not appear to be normal this semester in view of Covid19. to

maintain High school students will have until January 2021 to complete any outstanding classwork for grades

Under the new system, university teachers and their heads of department, will be asked to grade and rank their students. It is felt that as teachers know their students best, this is the most fair and accurate way of evaluation of a students performance across the course of their A-Levels.

Problem Statement

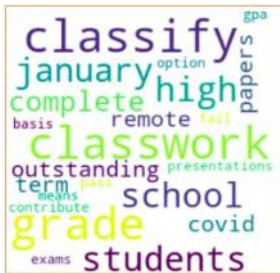
- **Input:** Covid-19 education grading new system
- **Task:** Identify the countries from the article.
- **Problem:** How do we represent?

The HRD ministry said that, In case the situation does not appear to be normal this semester in view of Covid19, to

maintain High school students will have until January 2021 to complete any outstanding classwork for grades

the board. Under the new system, university teachers and their heads of department, will be asked to grade and rank their students. It is felt that as teachers know their students best, this is the most fair and accurate way of evaluation of a student's performance across the course of their A-Levels.

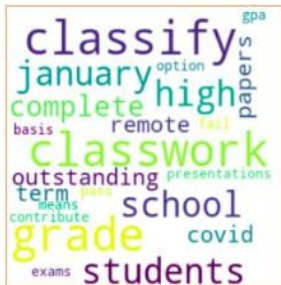
Which article?



Which article?



INDIA



USA



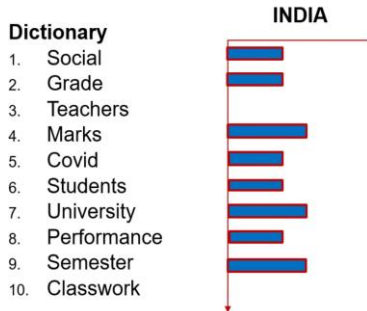
UK

Bag of Words Histogram

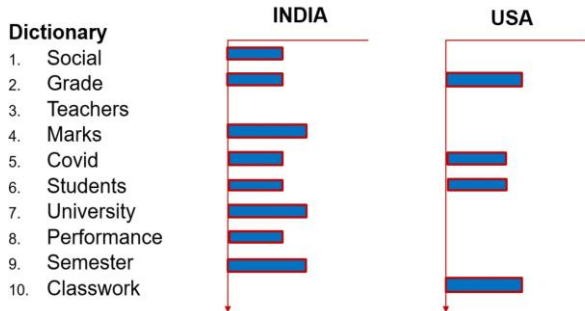
Dictionary

1. Social
2. Grade
3. Teachers
4. Marks
5. Covid
6. Students
7. University
8. Performance
9. Semester
10. Classwork

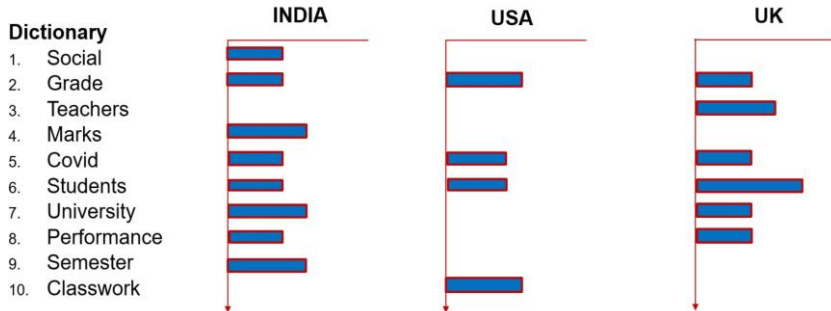
Bag of Words Histogram



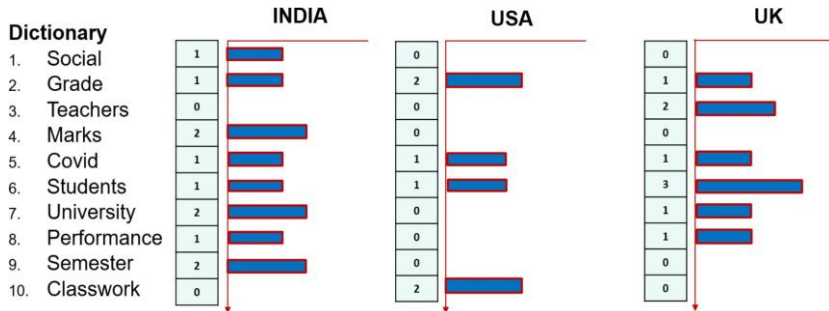
Bag of Words Histogram



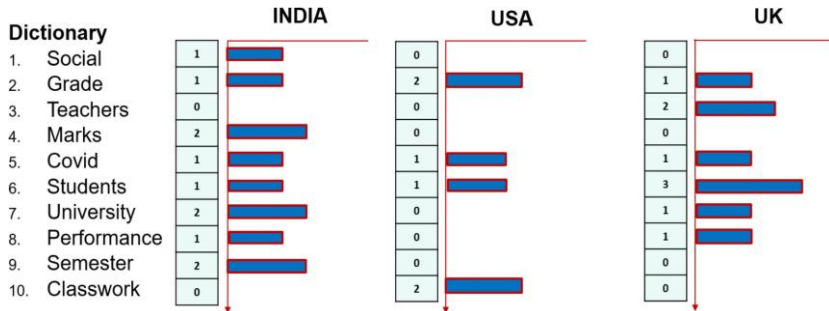
Bag of Words Histogram



Bag of Words Histogram

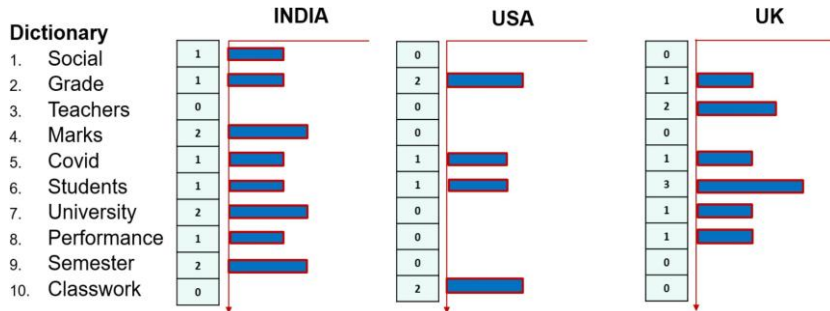


Bag of Words Histogram



- Orderless document representation; frequencies of words from a dictionary

Bag of Words Histogram



- Orderless document representation; frequencies of words from a dictionary
- Classification to determine document category

Histogram of Word Occurrences

Histogram of Word Occurrences

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

- ▶ Demo_Bag_of_Words_Python_Sckit_Learn
 -) Objective : To extract features using Python and Machine Learning with scikit-learn

1-Hot Representation of a Word

- **D1** : Andrew is a tall boy.
- **D2** : Ram is a good boy. Ratna is also good.

Dictionary

A
Also
Andrew
Boy
Good
Tall
Is
Ram
Ratna

1-Hot Representation of a Word

- **D1** : Andrew is a tall boy.
- **D2** : Ram is a good boy. Ratna is also good.

Dictionary	A	$[100000000]^T$
	Also	$[010000000]^T$
	Andrew	$[001000000]^T$
	Boy	$[000100000]^T$
	Good	$[000010000]^T$
	Tall	$[000001000]^T$
	Is	$[000000100]^T$
	Ram	$[000000010]^T$
	Ratna	$[000000001]^T$

1-Hot Representation of a Word

- **D1** : Andrew is a tall boy.
- **D2** : Ram is a good boy. Ratna is also good.

Dictionary	A	$[100000000]^T$	} 1-Hot representations
	Also	$[010000000]^T$	
	Andrew	$[001000000]^T$	
	Boy	$[000100000]^T$	
	Good	$[000010000]^T$	
	Tall	$[000001000]^T$	
	Is	$[000000100]^T$	
	Ram	$[000000010]^T$	
	Ratna	$[000000001]^T$	

Histogram vs. 1-Hot Representation

- Histogram is the sum of one hot representations of all the words
- **D1** : Andrew is a tall boy

A
Also
Andrew
Boy
Good
Tall
Is
Ram
Ratna

Histogram vs. 1-Hot Representation

- Histogram is the sum of one hot representations of all the words
- **D1 :** Andrew is a tall boy

A
Also
Andrew
Boy
Good
Tall
Is
Ram
Ratna

1-Hot Representation

0
0
1
0
0
0
0
0
0

Histogram vs. 1-Hot Representation

- Histogram is the sum of one hot representations of all the words

- **D1 :** Andrew is a tall boy

A
Also
Andrew
Boy
Good
Tall
Is
Ram
Ratna

1-Hot Representation

 $\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$

Histogram vs. 1-Hot Representation

- Histogram is the sum of one hot representations of all the words

➤ **D1 :** Andrew is a tall boy

A
Also
Andrew
Boy
Good
Tall
Is
Ram
Ratna

1-Hot Representation

Andrew	is	a	tall	boy
0	0	1	0	0
0	0	0	0	0
1	0	0	0	0
0	0	0	0	1
0	0	0	0	0
0	0	0	1	0
0	1	0	0	0
0	0	0	0	0
0	0	0	0	0

Histogram vs. 1-Hot Representation..

- Histogram is the sum of one hot representations of all the words

➤ **D1 :** Andrew is a tall boy

A
Also
Andrew
Boy
Good
Tall
Is
Ram
Ratna

1-Hot Representation

$$\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Histogram vs. 1-Hot Representation

- Histogram is the sum of one hot representations of all the words

➤ **D1 :** Andrew is a tall boy

A	0	0	1	0	0	1
Also	0	0	0	0	0	0
Andrew	1	0	0	0	0	1
Boy	0	0	0	0	1	1
Good	0	0	0	0	0	0
Tall	0	0	0	1	0	1
Is	0	1	0	0	0	1
Ram	0	0	0	0	0	0
Ratna	0	0	0	0	0	0

1-Hot Representation + + + + = Histogram (Bag of Words)

Document Histogram

Histogram for both the documents:

Dictionary	A	1	1
	Also	0	1
	Andrew	1	0
	Boy	1	1
	Good	0	2
	Tall	1	0
	Is	1	2
	Ram	0	1
	Ratna	0	1
		D1	D2

- **D1** : Andrew is a tall boy.
- **D2** : Ram is a good boy.
Ratna is also good.

Comments: Weight Words (TF-IDF)

- ▶ Not all words are equally useful.
 - › Some words do not add value.
 - › e.g., **the**, **of**, **and**, **is**, **a**, etc.
- ▶ **Stop Words:** removed from the dictionary
- ▶ Some words are more important
 - › Words that occur multiple times
 - › Words that are unique to a document

- ▶ Frequent Words (TF:Term Frequency)
 - › Higher the frequency, more relevant
 - › TF measures the frequency (count) of a term in a document
 - › TF is often divided by the document length

$$TF(T) = \frac{\# \text{ of occurrences of } T \text{ in } D_i}{\# \text{ of words in } D_i}$$

- ▶ Rare/Unique Words (IDF:Inverse Document Frequency)
 - › Need to weigh down the frequent terms while scale up the rare ones

$$IDF(T) = \log_e \left(\frac{\# \text{ of documents}}{\# \text{ of docs with term } T} \right)$$

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary		<i>A</i>	
		<i>Also</i>	
	☐	<i>Andrew</i>	☐
	☐	<i>Boy</i>	☐
		<i>Good</i>	
	☐	<i>Tall</i>	☐
	☐	<i>Is</i>	☐
	☐	<i>Ram</i>	☐
		<i>Ratna</i>	

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary		A	
		Also	
	☐	Andrew	☐
	☐	Boy	☐
	☐	Good	
	☐	Tall	☐
	☐	Is	☐
	☐	Ram	☐
		Ratna	

TF of "Andrew" in D_1 =

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary

	A	
	Also	
☐	Andrew	☐
☐	Boy	☐
☐	Good	☐
☐	Tall	☐
	Is	
	Ram	
	Ratna	

$$TF = \frac{\text{\# of occurrences of } T \text{ in } D_i}{\text{\# of words in } D_i}$$

TF of "Andrew" in D_1 =

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary		A	
		Also	
	☐	Andrew	☐
	☐	Boy	☐
	☐	Good	☐
	☐	Tall	☐
		Is	
		Ram	
		Ratna	

$$TF = \frac{\# \text{ of occurrences of } T \text{ in } D_i}{\# \text{ of words in } D_i}$$

$$TF \text{ of "Andrew" in } D_1 = \frac{1}{5} = 0.2$$

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary

	A	
	Also	
☐	Andrew	☐
☐	Boy	☐
☐	Good	☐
☐	Tall	☐
	Is	
	Ram	
	Ratna	

$$TF = \frac{\# \text{ of occurrences of } T \text{ in } D_i}{\# \text{ of words in } D_i}$$

$$TF \text{ of "Andrew" in } D_1 = \frac{1}{5} = 0.2$$

$$TF \text{ of "good" in } D_2 =$$

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary

	A	
	Also	
☐	Andrew	☐
☐	Boy	☐
☐	Good	☐
☐	Tall	☐
	Is	
	Ram	
	Ratna	

$$TF = \frac{\text{\# of occurrences of } T \text{ in } D_i}{\text{\# of words in } D_i}$$

$$TF \text{ of "Andrew" in } D_1 = \frac{1}{5} = 0.2$$

$$TF \text{ of "good" in } D_2 = \frac{2}{9} = 0.22$$

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary		<i>A</i>	
		<i>Also</i>	
	☐	<i>Andrew</i>	☐
	☐	<i>Boy</i>	☐
		<i>Good</i>	
	☐	<i>Tall</i>	☐
	☐	<i>Is</i>	☐
	☐	<i>Ram</i>	☐
		<i>Ratna</i>	

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary		A	
		Also	
	☐	Andrew	☐
		Boy	
	☐	Good	☐
		Tall	
	☐	Is	☐
		Ram	
		Ratna	

$IDF(rA)$

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary	☐	A	☐
	☐	Also	☐
	☐	Andrew	☐
	☐	Boy	☐
	☐	Good	☐
	☐	Tall	☐
	☐	Is	☐
		Ram	
		Ratna	

$$IDF(T) = \log_e \frac{\# \text{ of documents}}{\# \text{ of docs with term } T}$$

$$IDF(rA^r)$$

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary	☐	A	☐
	☐	Also	☐
	☐	Andrew	☐
	☐	Boy	☐
	☐	Good	☐
	☐	Tall	☐
	☐	Is	☐
		Ram	
		Ratna	

$$IDF(T) = \log_e \frac{\# \text{ of documents}}{\# \text{ of docs with term } T}$$

$$IDF('A') = \log_e \left(\frac{2}{2} \right) = 0$$

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary	☐	A	☐
	☐	Also	☐
	☐	Andrew	☐
	☐	Boy	☐
	☐	Good	☐
	☐	Tall	☐
	☐	Is	☐
		Ram	
		Ratna	

$$IDF(T) = \log_e \frac{\# \text{ of documents}}{\# \text{ of docs with term } T}$$

$$IDF(rA^r) = \log_e \left(\frac{2}{2} \right) = 0$$

$$IDF(rAndrew^r)$$

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary	☐	A	☐
	☐	Also	☐
	☐	Andrew	☐
	☐	Boy	☐
	☐	Good	☐
	☐	Tall	☐
	☐	Is	☐
		Ram	
		Ratna	

$$IDF(T) = \log_e \frac{\# \text{ of documents}}{\# \text{ of docs with term } T}$$

$$IDF('A') = \log_e \left(\frac{2}{2} \right) = 0$$

$$IDF('Andrew') = \log_e \left(\frac{2}{1} \right) = 0.69$$

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary

☐	A	☐
☐	Also	☐
☐	Andrew	☐
☐	Boy	☐
☐	Good	☐
☐	Tall	☐
☐	Is	☐
	Ram	
	Ratna	

$$IDF(T) = \log_e \frac{\# \text{ of documents}}{\# \text{ of docs with term } T}$$

$$IDF('A') = \log_e \left(\frac{2}{2} \right) = 0$$

$$IDF('Andrew') = \log_e \left(\frac{2}{1} \right) = 0.69$$

$$IDF('is')$$

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary

☐	A	☐
☐	Also	☐
☐	Andrew	☐
☐	Boy	☐
☐	Good	☐
☐	Tall	☐
☐	Is	☐
	Ram	
	Ratna	

$$IDF(T) = \log_e \frac{\# \text{ of documents}}{\# \text{ of docs with term } T}$$

$$IDF('A') = \log_e \left(\frac{2}{2} \right) = 0$$

$$IDF('Andrew') = \log_e \left(\frac{2}{1} \right) = 0.69$$

$$IDF('is') = \log_e \left(\frac{2}{2} \right) = 0$$

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary

☐	A	☐
☐	Also	☐
☐	Andrew	☐
☐	Boy	☐
☐	Good	☐
☐	Tall	☐
☐	Is	☐
	Ram	
	Ratna	

$$IDF(T) = \log_e \frac{\# \text{ of documents}}{\# \text{ of docs with term } T}$$

$$IDF('A') = \log_e \left(\frac{2}{2} \right) = 0$$

$$IDF('Andrew') = \log_e \left(\frac{2}{1} \right) = 0.69$$

$$IDF('is') = \log_e \left(\frac{2}{2} \right) = 0$$

$$IDF('good')$$

D_1 : Andrew is a tall boy.

D_2 : Ram is a good boy. Ratna is also good.

Dictionary

☐	A	☐
☐	Also	☐
☐	Andrew	☐
☐	Boy	☐
☐	Good	☐
☐	Tall	☐
☐	Is	☐
	Ram	
	Ratna	

$$IDF(T) = \log_e \frac{\# \text{ of documents}}{\# \text{ of docs with term } T}$$

$$IDF(rA^r) = \log_e \left(\frac{2}{2} \right) = 0$$

$$IDF(rAndrew^r) = \log_e \left(\frac{2}{1} \right) = 0.69$$

$$IDF(ris^r) = \log_e \left(\frac{2}{2} \right) = 0$$

$$IDF(rgood^r) = \log_e \left(\frac{2}{1} \right) = 0.69$$

- **D1** : Andrew is a tall boy.
- **D2** : Ram is a good boy. Ratna is also good.

Dictionary	A	0	0.2	0
	Also	0.69	0	0
	Andrew	0.69	0.2	0.14
	Boy	0	0.2	0
	Good	0.69	0	0
	Tall	0.69	0.2	0.14
	Is	0	0.2	0
	Ram	0.69	0	0
	Ratna	0.69	0	0

- **D1** : Andrew is a tall boy.
- **D2** : Ram is a good boy. Ratna is also good.

Dictionary	A	0	0.2	0
	Also	0.69	0	0
	Andrew	0.69	0.2	0.14
	Boy	0	0.2	0
	Good	0.69	0	0
	Tall	0.69	0.2	0.14
	Is	0	0.2	0
	Ram	0.69	0	0
	Ratna	0.69	0	0
		IDF		

- **D1** : Andrew is a tall boy.
- **D2** : Ram is a good boy. Ratna is also good.

$$\begin{array}{c} \text{Dictionary} \end{array}
 \begin{bmatrix} \text{A} \\ \text{Also} \\ \text{Andrew} \\ \text{Boy} \\ \text{Good} \\ \text{Tall} \\ \text{Is} \\ \text{Ram} \\ \text{Ratna} \end{bmatrix}
 \begin{bmatrix} 0 \\ 0.69 \\ 0.69 \\ 0 \\ 0.69 \\ 0.69 \\ 0 \\ 0.69 \\ 0.69 \end{bmatrix}
 \times
 \begin{bmatrix} 0.2 \\ 0 \\ 0.2 \\ 0.2 \\ 0 \\ 0.2 \\ 0.2 \\ 0 \\ 0 \end{bmatrix}
 \begin{bmatrix} 0 \\ 0 \\ 0.14 \\ 0 \\ 0 \\ 0.14 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

IDF

TF-IDF

- **D1** : Andrew is a tall boy.
- **D2** : Ram is a good boy. Ratna is also good.

Dictionary	A	0	0.2	0
	Also	0.69	0	0
	Andrew	0.69	0.2	0.14
	Boy	0	0.2	0
	Good	0.69	0	0
	Tall	0.69	0.2	0.14
	Is	0	0.2	0
	Ram	0.69	0	0
	Ratna	0.69	0	0
		IDF	TFIDF	

$\times$

- **D1** : Andrew is a tall boy.
- **D2** : Ram is a good boy. Ratna is also good.

$$\begin{array}{c} \text{Dictionary} \end{array}
 \begin{bmatrix} \text{A} \\ \text{Also} \\ \text{Andrew} \\ \text{Boy} \\ \text{Good} \\ \text{Tall} \\ \text{Is} \\ \text{Ram} \\ \text{Ratna} \end{bmatrix}
 \begin{bmatrix} 0 \\ 0.69 \\ 0.69 \\ 0 \\ 0.69 \\ 0.69 \\ 0 \\ 0.69 \\ 0.69 \end{bmatrix}
 \begin{matrix} \\ \\ \times \\ \\ \\ \\ \\ \\ \end{matrix}
 \begin{bmatrix} 0.2 \\ 0 \\ 0.2 \\ 0.2 \\ 0 \\ 0.2 \\ 0.2 \\ 0 \\ 0 \end{bmatrix}
 =
 \begin{bmatrix} 0 \\ 0 \\ 0.14 \\ 0 \\ 0 \\ 0.14 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

IDF
TFD1

TF-IDF

- **D1** : Andrew is a tall boy.
- **D2** : Ram is a good boy. Ratna is also good.

$$\begin{array}{c} \text{Dictionary} \\ \left[\begin{array}{c} A \\ \text{Also} \\ \text{Andrew} \\ \text{Boy} \\ \text{Good} \\ \text{Tall} \\ \text{Is} \\ \text{Ram} \\ \text{Ratna} \end{array} \right] \end{array} \times \begin{array}{c} \left[\begin{array}{c} 0 \\ 0.69 \\ 0.69 \\ 0 \\ 0.69 \\ 0.69 \\ 0 \\ 0.69 \\ 0.69 \end{array} \right] \\ \text{IDF} \end{array} = \begin{array}{c} \left[\begin{array}{c} 0 \\ 0 \\ 0.14 \\ 0 \\ 0 \\ 0.14 \\ 0 \\ 0 \\ 0 \end{array} \right] \\ \text{TFIDF} \\ \text{D1} \end{array}$$

The diagram illustrates the calculation of TFIDF for document D1. It shows a Dictionary vector multiplied by an IDF vector to produce a TFIDF vector for D1. The IDF vector contains values for each word in the dictionary, and the TFIDF vector for D1 contains the resulting values for each word in D1.

TF-IDF

- **D1** : Andrew is a tall boy.
- **D2** : Ram is a good boy. Ratna is also good.

$$\begin{array}{c} \text{Dictionary} \\ \left[\begin{array}{c} A \\ \text{Also} \\ \text{Andrew} \\ \text{Boy} \\ \text{Good} \\ \text{Tall} \\ \text{Is} \\ \text{Ram} \\ \text{Ratna} \end{array} \right] \end{array} \times \begin{array}{c} \left[\begin{array}{c} 0 \\ 0.69 \\ 0.69 \\ 0 \\ 0.69 \\ 0.69 \\ 0 \\ 0.69 \\ 0.69 \end{array} \right] \\ \text{IDF} \end{array} = \begin{array}{c} \left[\begin{array}{c} 0.11 \\ 0.11 \\ 0 \\ 0.11 \\ 0.22 \\ 0 \\ 0.22 \\ 0.11 \\ 0.11 \end{array} \right] \\ \text{TFD2} \end{array} = \begin{array}{c} \left[\begin{array}{c} 0 \\ 0.08 \\ 0 \\ 0.0 \\ 0.15 \\ 0 \\ 0 \\ 0.08 \\ 0.08 \end{array} \right] \\ \text{TFIDF} \\ \text{D2} \end{array}$$

► Actual text:

Word Count = 89

In reality, the laborer belongs to capital before he has sold himself to capital. His economic bondage is both brought about and concealed by the periodic sale of himself, by his change of masters, and by the oscillation in the market price of labor power. Capitalist production, therefore, under its aspect of a continuous connected process, of a process of reproduction, produces not only commodities, not only surplus value, but it also produces and reproduces the capitalist relation; on the one side the capitalist, on the other the wage-laborer.

► Actual text:

Word Count = 89

In reality, the laborer belongs to capital before he has sold himself to capital. His economic bondage is both brought about and concealed by the periodic sale of himself, by his change of masters, and by the oscillation in the market price of labor power. Capitalist production, therefore, under its aspect of a continuous connected process, of a process of reproduction, produces not only commodities, not only surplus value, but it also produces and reproduces the capitalist relation; on the one side the capitalist, on the other the wage-laborer.

Stopwords: the, a, and, is, of, to, in, on, by, he, his, has, it, its, therefore, under, side, one, not, only, but, also, other

Stopwords..

- After removing Stop words

Word Count = 37

reality, laborer, belongs, capital, sold, capital, economic, bondage, brought, concealed, periodic, sale, change, masters, oscillation, market, price, labor, power, capitalist, production, aspect, continuous, connected, process, process, reproduction, produces, commodities, surplus, value, produces, reproduces, capitalist, relation, capitalist, wage-laborer.

Stopwords..

► After removing Stop words

Word Count = 37

reality, laborer, belongs, capital, sold, capital, economic, bondage, brought, concealed, periodic, sale, change, masters, oscillation, market, price, labor, power, capitalist, production, aspect, continuous, connected, process, process, reproduction, produces, commodities, surplus, value, produces, reproduces, capitalist, relation, capitalist, wage-laborer.

Unigrams

- | | | |
|--------------------|---------------|-----------------|
| ► Capital(ist):5 | ► Economic:1 | ► Periodic:1 |
| ► Labor(er):3 | ► Reality:1 | ► Change:1 |
| ► (Re)produces:3 | ► Belongs:1 | ► Masters:1 |
| ► (Re)production:2 | ► Bondage:1 | ► Oscillation:1 |
| ► Process:2 | ► Brought:1 | ► Market:1 |
| ► Sold/sale:2 | ► Concealed:1 | ► Price:1 |

N-Grams: Capturing Context

- ▶ An n-gram is a sequence of n consecutive items (words, characters)
 - › Unigram, Bigram, Trigram, 4-gram, 5-gram, etc.
- ▶ Histogram or BoW was counting unigrams.
- ▶ What about Bigrams or Trigrams?

Bigrams

Friends, Romans, countrymen, lend me your ears
I come to bury Caesar, not to praise him.
The evil that men do lives after them;
The good is oft interred with their bones;
So let it be with Caesar.
The noble Brutus hath told you Caesar was ambitious:
If it were so, it was a grievous fault,
And grievously hath Caesar answered it.
Here, under leave of Brutus and the rest
For Brutus is an honorable man;
So are they all, all honorable men
Come I to speak in Caesar's funeral.
He was my friend, faithful and just to me:
But Brutus says he was ambitious;
And Brutus is an honorable man.
He hath brought many captives home to Rome
Whose ransoms did the general coffers fill:
Did this in Caesar seem ambitious?
When that the poor have cried, Caesar hath wept:
Ambition should be made of sterner stuff:

.....

Friends, Romans, countrymen, lend me your ears
I come to bury Caesar, not to praise him.
The evil that men do lives after them;
The good is oft interred with their bones;
So let it be with Caesar.
The noble Brutus hath told you Caesar was ambitious:
If it were so, it was a grievous fault,
And grievously hath Caesar answered it.
Here, under leave of Brutus and the rest
For Brutus is an honorable man;
So are they all, all honorable men
Come I to speak in Caesar's funeral.
He was my friend, faithful and just to me:
But Brutus says he was ambitious;
And Brutus is an honorable man.
He hath brought many captives home to Rome
Whose ransoms did the general coffers fill:
Did this in Caesar seem ambitious?
When that the poor have cried, Caesar hath wept:
Ambition should be made of sterner stuff:

.....



Marc Antony
Caesars funeral
WILLIAM SHAKESPEARE

Bigrams

Friends, Romans, countrymen, lend me your ears
I come to bury Caesar, not to praise him.
The evil that men do lives after them;
The good is oft interred with their bones;
So let it be with Caesar.
The noble Brutus hath told you Caesar was ambitious:
If it were so, it was a grievous fault,
And grievously hath Caesar answered it.
Here, under leave of Brutus and the rest
For Brutus is an honorable man;
So are they all, all honorable men
Come I to speak in Caesar's funeral.
He was my friend, faithful and just to me:
But Brutus says he was ambitious;
And Brutus is an honorable man.
He hath brought many captives home to Rome
Whose ransoms did the general coffers fill:
Did this in Caesar seem ambitious?
When that the poor have cried, Caesar hath wept:
Ambition should be made of sterner stuff:
.....

- ('of', 'caesar'):6
- ('you', 'know'):6
- ('you', 'all'):5
- ('was', 'ambitious'):4
- ('brutus', 'is'):4
- ('is', 'an'):4
- ('an', 'honorable'):4
- ('honorable', 'man'):4
- ('honorable', 'men'):4
- ('he', 'was'):4
- ('in', 'his'):4
- ('know', 'not'):4
- ('to', 'speak'):3
- ('brutus', 'says'):3
- ('ambitious', 'and'):3
- ('and', 'brutus'):3
- ('he', 'hath'):3
- ('tell', 'you'):3
- ('let', 'it'):2
- ('with', 'caesar'):2
- ('the', 'noble'):2

Experiment

- ▶ Demo_N_Grams
 -) Objective : n-gram representation

Use of N-grams

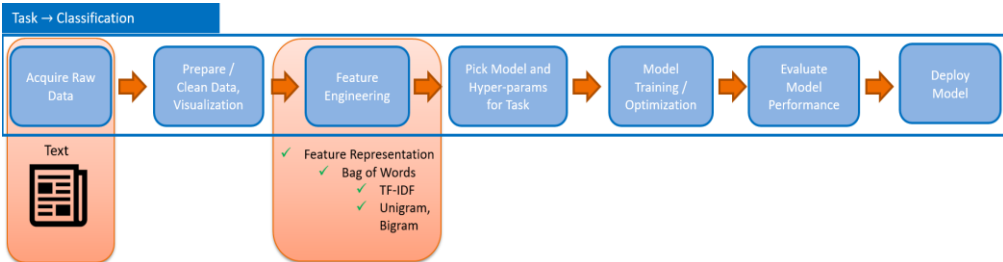
- Representing sentences, documents
- Modeling language
 - › Predicting next word (Auto-complete)
 - › Resolving ambiguity (Speech recognition, OCR)
 - › Machine Translation (choosing one sentence over another)
 - › Many other tasks: Question answering, classification, etc.
- Smoothing N-grams

- ▶ Standard Techniques:
 - › Remove stop words; Non-informative words
 - › Weigh words differently
- ▶ Histogram (Bag of Words)
 - › Unordered simple representation
 - › Though the structure of sentences lost in histogram, we can still classify based on ordering of words.

Summary

- ▶ Deeper Techniques from NLP:
 - › N-gram Representations
 - › Identify named entities (proper nouns), word morphology.
- ▶ Learning Representations

Summary



Thanks!!

Questions?